

Tien-Chen Lin

*Cytoskeletal
neurobiology ·
Mechanobiology
of brain
development*

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Research summary

My research lies at the intersection of molecular neurobiology, cytoskeletal dynamics, and mechanotransduction. I study how diverse cell types coordinate their morphological changes during brain development, combining theoretical modelling of cytoskeletal dynamics with live imaging, optogenetic manipulation *in vitro* and *in vivo*, biochemical reconstitution, and large-scale omics. My long-term goal is to uncover the biophysical principles governing the self-organization of the nervous system, and their dysregulation in aging and degeneration.

Education

2009–2014 **Ph.D., Natural Sciences (Dr. rer. nat., *magna cum laude*)**, University of Heidelberg, Germany

2005–2006 **M.Sc., Biochemistry & Molecular Biology**, National Yang-Ming University, Taiwan

2002–2005 **B.Sc., Life Sciences**, National Yang-Ming University, Taiwan

Research experience

2017–present **Postdoctoral Researcher**, DZNE, Bonn — Axon Growth & Regeneration (Prof. F. Bradke)
Discovered a soma-based, Arp2/3–actomyosin local-excitation/global-inhibition mechanism of CNS neuronal polarization; reconciled the roles of actin networks in neurite growth; built an R/ImageJ analysis pipeline for neurite-growth dynamics; collaborative omics of axon regeneration, reactive astrocytes after stroke, and DRG-neuron subtypes.

2014–2017 **Postdoctoral Researcher**, ZMBH, University of Heidelberg (Prof. E. Schiebel)
Established MOZART1 as an integral part of the γ -tubulin small complex in *Candida albicans* and human cells, and as a driver of its assembly into a nucleation-competent ring; mapped conserved interactions among MOZART1, γ -TuSC, and the receptors pericentrin and CDK5RAP2.

2009–2014 **Doctoral Researcher**, ZMBH, University of Heidelberg (Prof. E. Schiebel)
Elucidated the mitotic phospho-regulation of the γ -tubulin small complex and pericentrin; characterized γ -TuSC–receptor interactions and the divergent targeting of γ -tubulin complexes to microtubule organizing centers.

Honors & fellowships

2016 **EMBO Short-term Fellowship**

2012 **LGFG Fellowship** — “Regulation of Cell Division”

Selected publications

2026 **Lin, T.-C.**, Coles, C.H., Alfadil, E., *et al.*, Bradke, F. An intrinsic cytoskeletal oscillator establishes neuronal polarity. *Nature* (2026). doi:10.1038/s41586-026-10755-6

- 2016 **Lin, T.-C.**, Neuner, A., Flemming, D., *et al.* MOZART1 and γ -tubulin complex receptors are both required to turn γ -TuSC into an active microtubule nucleation template. *J. Cell Biol.* 215(6), 823–840.
- 2015 **Lin, T.-C.**, Neuner, A., Schiebel, E. Targeting of γ -tubulin complexes to microtubule organizing centers: conservation and divergence. *Trends Cell Biol.* 25(5), 296–307.
- 2014 **Lin, T.-C.**, Neuner, A., Schlosser, Y.T., *et al.* Cell-cycle dependent phosphorylation of yeast pericentrin regulates γ -TuSC-mediated microtubule nucleation. *eLife* 3, e02208.
- 2011 **Lin, T.-C.**, Gombos, L., Neuner, A., *et al.* Phosphorylation of the yeast γ -tubulin Tub4 regulates microtubule function. *PLoS ONE* 6(5), e19700.
- Full list: darkbreaker0.github.io/publications*

Invited talks (selected)

- 2023 **Microtubules in Neurons (Chiemsee); The Cytoskeleton of Neurons and Glia (webinar series)**
- 2022 **CSHL “Molecular Mechanisms of Neuronal Connectivity” (New York); FENS Symposium (Paris); EMBO Workshop (Taipei); invited lecture with F. Bradke, NYCU College of Life Sciences (Taiwan)**

Teaching & service

- 2020–2023 **Coordinator & lecturer**, *M.Sc. “Molecular Cell Biology”, University of Bonn*
Light-microscopy basics, live-cell imaging, and data visualization with R.
- 2019–present **Associate Faculty Member**, *F1000 / Faculty Opinions*
- Mentoring Supervision of M.Sc. research projects and rotation students.

Professional development

- EMBO Laboratory Leadership for Group Leaders; Project Management for Scientists; Negotiation for Scientists.
- Courses Advanced Techniques in Molecular Neuroscience (CSHL); Super-Resolution Microscopy and Transcriptome/Proteome Integration (EMBL).